AP PRECALCULUS

Unit 1 • Worksheet 1.4A

Worksheet 1.4A • Polynomial Functions and Rates of Change

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the greatest possible number of local extrema of a polynomial of degree 5?

(A) 5
(B) 4
(C) 3
(D) 6

Question 2

NO CALCULATOR

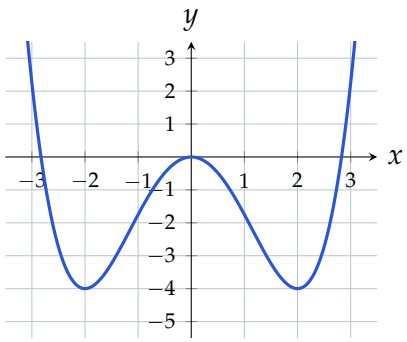
What is the degree and leading coefficient of $p(x) = -3x(x+2)^2(1-x)$?

(A) degree 4, leading coefficient -3
(B) degree 4, leading coefficient 3
(C) degree 3, leading coefficient 3
(D) degree 5, leading coefficient -3

Question 3

NO CALCULATOR

The graph of $p(x) = \frac{1}{4}x^4 - 2x^2$ is shown. What is the absolute minimum value of p?



(A) 0
(B) -2
(C) -5
(D) -4

Question 4

NO CALCULATOR

Using the same graph, what is the absolute maximum of p on $[-1, 3]$?

(A) 0
(B) 2.25
(C) -1.75
(D) 4

Question 5 (Free response)

NO CALCULATOR

The polynomial $p(x) = x^3 - 6x^2 + 9x$ has a local maximum at $(1, 4)$, a local minimum at $(3, 0)$, and a point of inflection at $(2, 2)$.

(a) On which intervals is p increasing? Decreasing?
(b) Find the absolute maximum and minimum values of p on $[0, 4]$.
(c) On which interval is p decreasing and concave up? Describe how p changes there.

Question 6 (Free response)

NO CALCULATOR

Sketch the graph of a polynomial of degree 4 with a positive leading coefficient, local minima at $(-2, -3)$ and $(2, -3)$, and a local maximum at $(0, 1)$. How many points of inflection must it have?